

SH1 Christchurch Motorway Speed Limit Change Impact

Before-After Evaluation: 100 → 110 km/h | Change Date: April 13, 2025

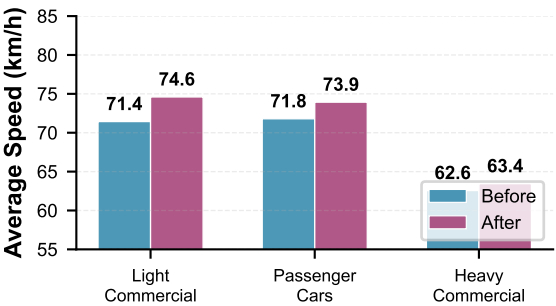
1,127
Motorway
Trips

135K
GPS
Points

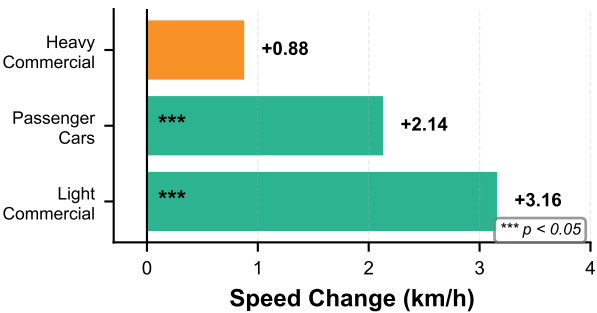
26
Crashes
Analyzed

+0.74
Overall Speed
Change (km/h)

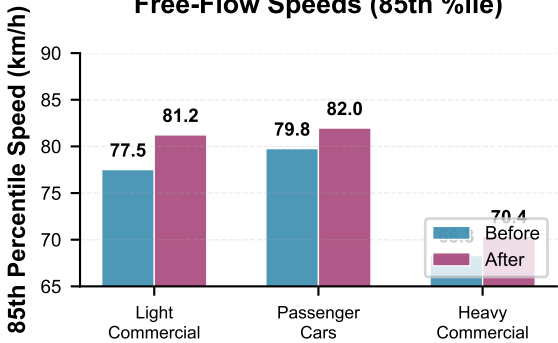
Mean Speed by Vehicle Type



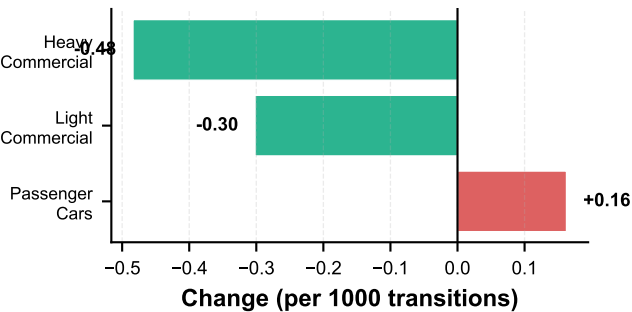
Magnitude of Change



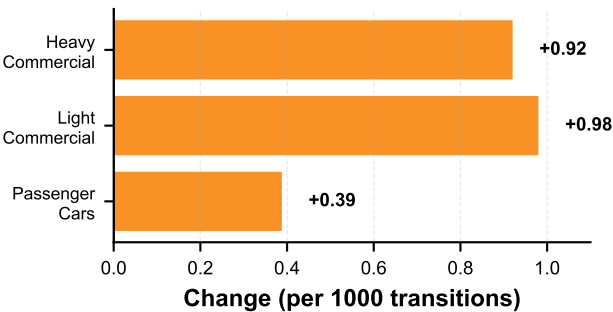
Free-Flow Speeds (85th %ile)



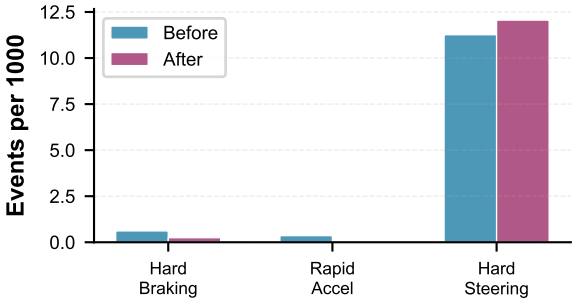
Hard Braking Changes



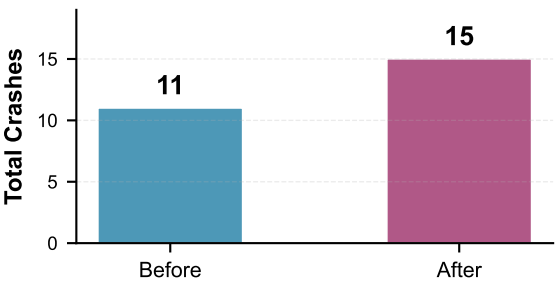
Hard Steering (All)



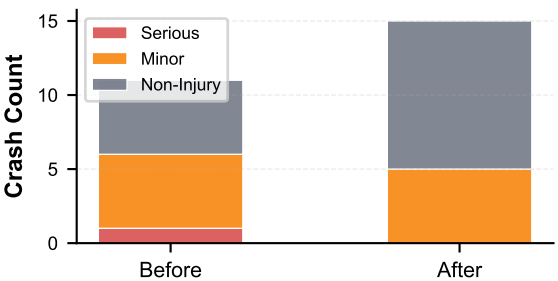
Overall Behavioral Metrics



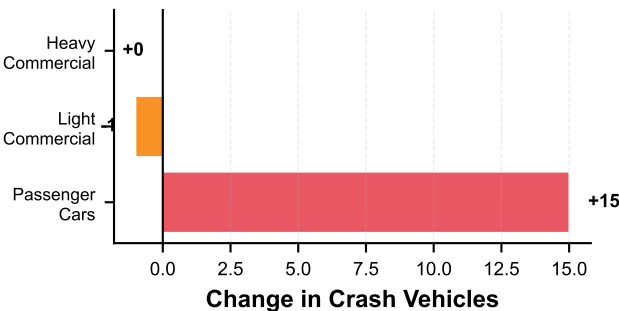
Crash Frequency



Crash Severity Distribution



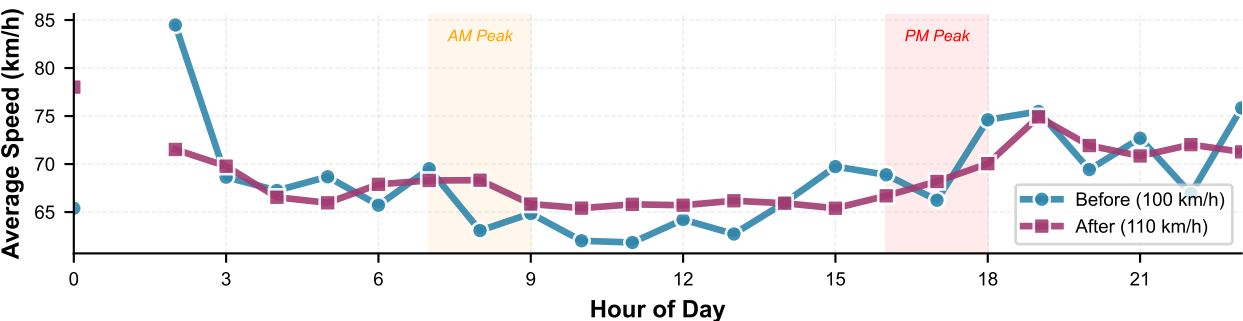
Crash Involvement



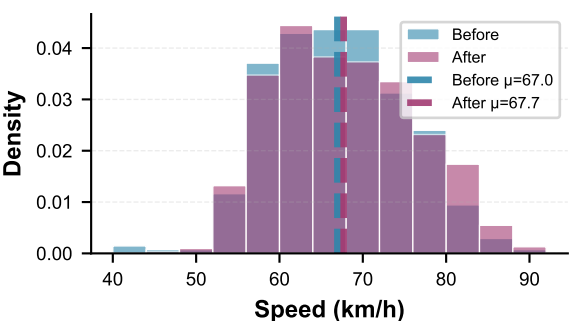
Temporal & Behavioral Patterns

Detailed Analysis of Speed Changes and Driving Behavior

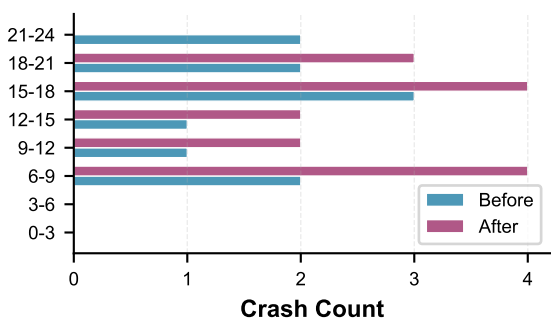
Speed Variation Throughout Day



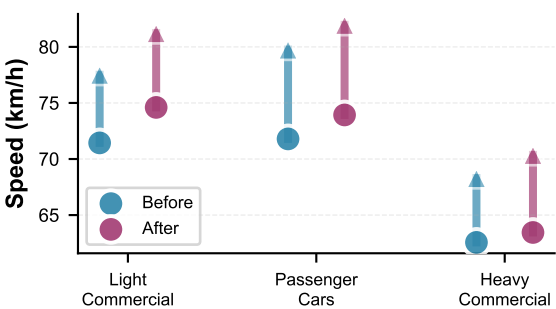
Speed Distribution



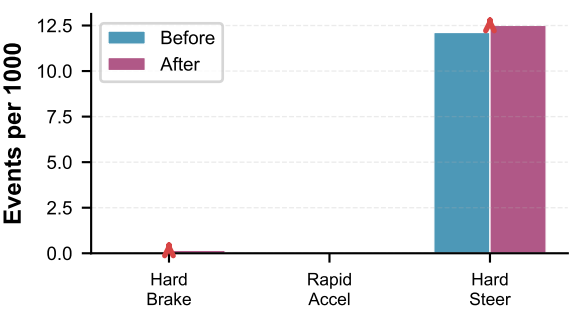
Crashes by Time of Day



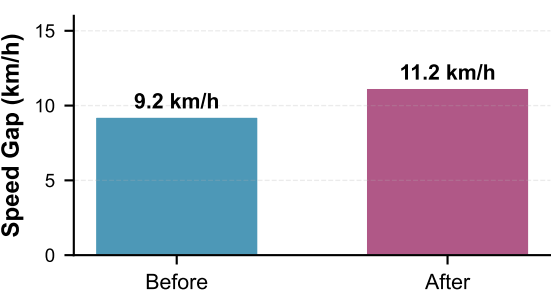
Speed Range (Mean & 85th)



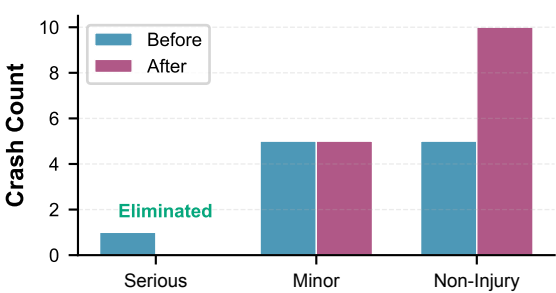
Passenger Car Behavior



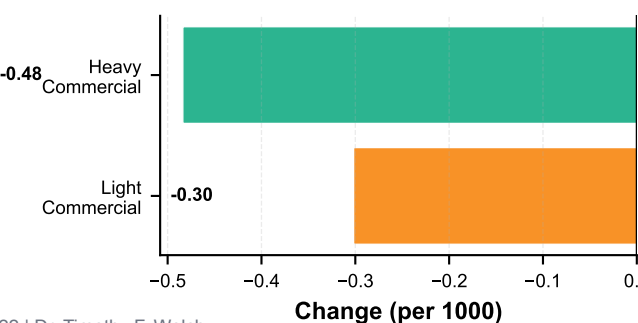
Vehicle Type Speed Gap



Crash Severity Changes



Commercial Hard Braking



Methodology & Study Area

Data Processing Pipeline & Geographic Analysis

Data Processing Pipeline

Phase 1 Validation 92,456 trips	Phase 2 Storage Parquet format	Phase 3 Geographic Filter Corridor bounds	Phase 4 Point Expansion 11.4M GPS points	Phase 5 Temporal Classify Before/After	Phase 6 Spatial Filter 1,127 trips
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Spatial Filter: 50m buffer from motorway | Perpendicular distance algorithm | $\geq 50\%$ points on motorway required

Study Area: SH1 Christchurch Motorway (Addison to Rolleston)



Statistical Validation Summary

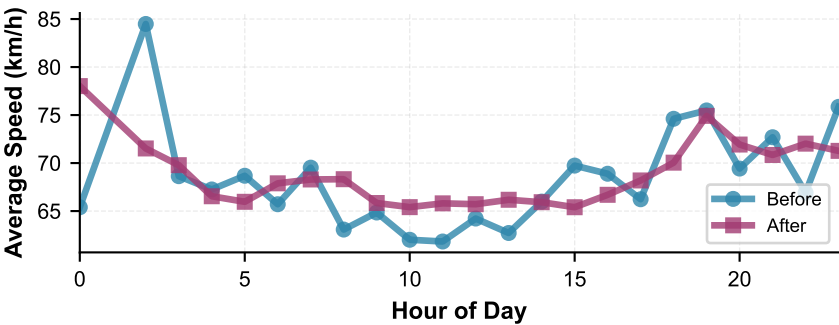
Vehicle Type	Sample (Before/After)	Mean Speed Change	p-value	Effect Size	Significance
LCV	127/244	+3.16 km/h	0.0000	0.45 (Small)	Significant
CAR	41/58	+2.14 km/h	0.0334	0.23 (Small)	Significant
HCV	173/478	+0.88 km/h	0.0733	0.13 (Small)	Not Sig.

Mann-Whitney U test (non-parametric) | $\alpha = 0.05$ | Cohen's d for effect size

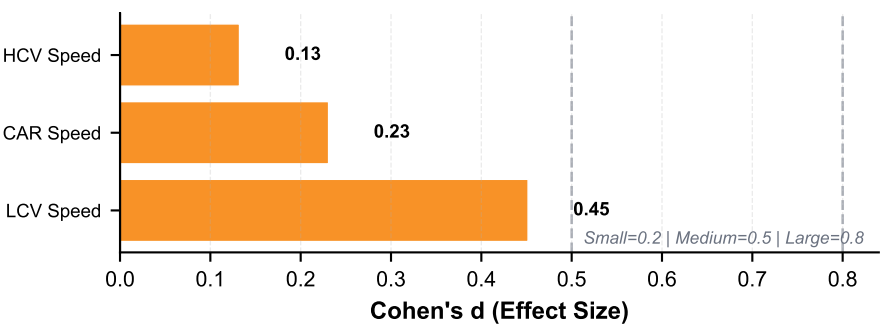
Temporal Patterns & Key Findings Summary

Time-Series Analysis and Comprehensive Findings

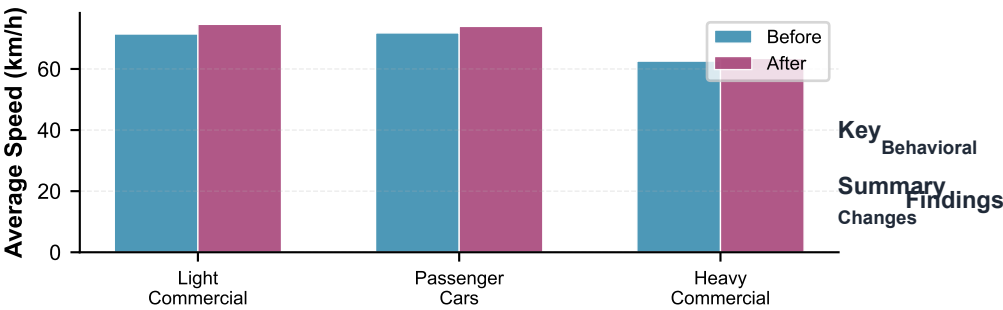
Speed Patterns by Hour



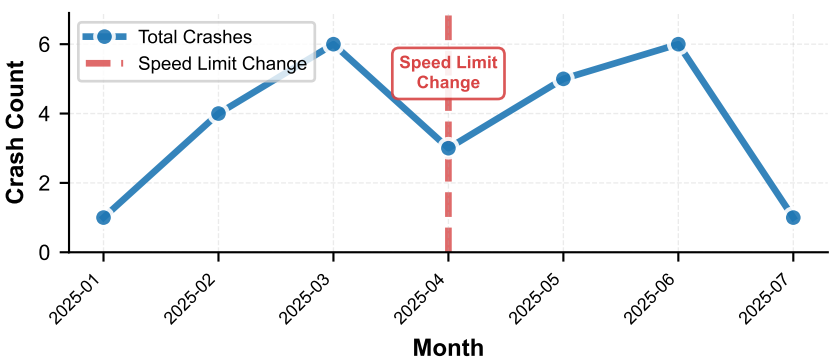
Statistical Effect Sizes



Speed Differential by Vehicle Type



Crash Frequency Over Time



Speed Impact

LCV: +3.16 km/h (+4.4%)
CAR: +2.14 km/h (+3.0%)
HCV: +0.88 km/h (+1.4%)
Overall: +0.74 km/h (+1.1%)

Behavioral Changes

Hard braking: -60% overall
CAR braking: +0.16 (worsened)
Hard steering: +7% all types
Rapid accel: -89% overall

Safety Outcomes

Total crashes: 11 → 15 (+36%)
CAR crashes: 13 → 28 (+115%)
Serious: 1 → 0 (none recorded)
Rear-end: 4 → 7 (+75%)